❄ Snowflake Learning Series – Zero-Copy Cloning in Snowflake ❄

What is Zero-Copy Cloning?

Zero-Copy Cloning in Snowflake allows users to create an **instant copy (snapshot) of a table, schema, or database** without actually duplicating the underlying data.

Instead of copying the data physically, Snowflake only copies the **metadata of the micro-partitions**.

Because of this design:

• Cloning is **extremely fast**
• It **does not duplicate data initially**
• It **does not consume additional storage at the time of cloning**

Key Concept Behind Zero-Copy Cloning

Snowflake stores table data in **micro-partitions**.

When a clone is created:

1. Snowflake **creates a new metadata entry**
2. The cloned object **references the same micro-partitions** as the original table
3. No physical data copy happens

This is why it is called **Zero-Copy Cloning**.

Example

Original table:

sales_table

Clone command:

```
CREATE TABLE sales_clone
CLONE sales_table;
```

Result:

• sales_clone is created instantly
• Both tables reference the **same micro-partitions**

No extra storage is used initially.

Storage Behavior After Cloning

Initially:

• Both tables share the same micro-partitions
• No additional storage cost

If data is modified in the clone:

Snowflake creates **new micro-partitions only for the changed data**.

Example:

If you update a row in the cloned table:

1. New micro-partition is created
2. Updated data is stored there
3. Original table remains unchanged

This ensures **data independence between original and cloned objects**.

Advantages of Zero-Copy Cloning

• Instant cloning of large datasets
• No storage cost initially
• Safe testing and development
• Independent modification of cloned objects
• Very useful for Dev, QA, and testing environments

Real-World Use Case

In a production environment:

Production Database → PROD_DB

Developers need data for testing.

Instead of copying huge datasets:

CREATE DATABASE DEV_DB CLONE PROD_DB;

Now developers can:

• Test queries
• Modify data
• Run experiments

Without affecting the production database.

Objects Supported for Cloning

Snowflake allows cloning of:

• Tables
• Schemas
• Databases

Example commands:

Clone Table

CREATE TABLE new_table CLONE old_table;

Clone Schema

CREATE SCHEMA new_schema CLONE old_schema;

Clone Database

CREATE DATABASE new_db CLONE old_db;

Important Cloning Considerations

Some objects are **not cloned completely**.

1️⃣ **Privileges**

Privileges are **not cloned**.

Access permissions must be assigned again.

2️⃣ **Stages**

• External stages → cloned
• Named internal stages → NOT cloned
• Table stages → cloned

3️⃣ **Pipes**

• Pipes referencing **external stages** → cloned
• Pipes referencing **internal stages** → NOT cloned

4️⃣ **Data History**

Data history **is not cloned**.

Features like **Time Travel history remain only with the original object**.

How Snowflake Implements Zero-Copy Cloning

Snowflake uses **metadata from the Cloud Services Layer**.

It does NOT use:

• Virtual Warehouse SSD Cache
• Query Cache

Instead, it updates **metadata references to micro-partitions**.

Example Command Explanation

Command:

CREATE OR REPLACE TABLE newTable CLONE table1;

What happens internally?

Snowflake does NOT copy data.

Instead:

• A new metadata entry is created
• Existing micro-partitions from table1 are mapped to newTable

This allows instant cloning.

Important Interview Question

Question

A point-in-time snapshot of data that can be updated by users is called:

1. Time Travel
2. Fail-Safe
3. Zero-Copy Cloning

Answer:

Zero-Copy Cloning

Question

Which Snowflake component enables Zero-Copy Cloning?

1. SSD Cache of Virtual Warehouse
2. Cache Layer
3. Metadata from Service Layer

Answer:

Metadata from Service Layer

Zero-Copy Cloning vs Time Travel

Feature	Zero-Copy Cloning	Time Travel
Purpose	Create independent copy	Access historical data
Storage	No initial storage	Uses historical storage
Modification	Allowed	Read historical data

Key Interview Points

Zero-Copy Cloning:

• Creates instant snapshot
• Copies metadata, not data
• Uses micro-partition references
• Initially consumes no storage
• Additional storage used only when data changes

Used mainly for:

• Development environments
• Testing environments
• Data backup scenarios

Quick Summary

Zero-Copy Cloning allows Snowflake users to **create fast, storage-efficient copies of databases, schemas, or tables** by sharing existing micro-partitions through metadata references.

This feature makes Snowflake extremely powerful for **development, testing, and analytics workflows**.